

Lecture No. 7

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List of Topic to be Discuss in this lecture

- Planning cost
- Estimation of costs
- Budget determination
- Controlling Cost

Knowledge Area -Cost Management

Knowledge Area	Process				
	Initiating	Planning	Executing	Monitoring & Control	Closing
Cost Management		Planning cost Estimation of costs Budget determination		Controlling costs	

What is Project Cost Management?

The process involved in **estimating, budgeting and controlling cost** so that the project can be completed within **approved budget**.

Cost Management Processes:

1. Planning cost
2. Estimation of costs
3. Budget determination
4. Controlling Cost

Planning Cost Process:

The Plan Cost Management process is where you plan out all the work you'll do to make sure your project doesn't cost more than you've budgeted.

Planning Cost - Tools and Techniques

- **Payback Period**
- **Net Present Value**
- **Internal Rate of Investment**

We have discussed in lecture No. 2 (Project Evaluation Techniques).

Estimation of Cost Process

The process of developing approximation of the monetary resources needed to complete project activities

Estimation of Cost - Tools and Techniques

Budget Estimate

Budget estimates are typically created based on historical information. If you've done a similar project in the past, you can use this historical information to predict what your current project is going to cost. Using historical information is called *analogous estimating*.

Estimation of Cost Process

Definitive Estimate

- Because so much of software development is based on labor, you may find that you're building your project estimates using a *parametric model*. A parametric model uses specific parameters, such as cost per hour, cost per line of code, or cost per network drop, to predict how much the project will cost.
- Parametric modeling is fine as long as you document how the estimate was created. In some organizations, parametric modeling won't work, because each developer's time is accounted for and billed independently.

Budget Determination Process:

Process of **aggregating** the estimated cost of individual activities or work packages to established an **authorized cost base line**.

Budget Determination Process - Tools and Techniques

Cost aggregation

You take your activity estimates and roll them up into control accounts on your work breakdown structure. That makes it easy for you to know what each work package in your project is going to cost. This tool is called **cost aggregation**.

Reserve analysis

When you evaluate the risks to your project, you will set aside some cash reserves to deal with any issues that might come your way. This tool is called **reserve analysis**.

Some Important Terminology:

Type of Reserves

There are two type of reserve

- ❑ **Contingency reserves** are used for addressing the known risks which remain on the project.
- ❑ **Management reserves** are used for addressing the unknown risks on the project.
- ❑ Contingency reserves are added to the project estimate to come up with the cost baseline.
- ❑ Management reserves are not a part of the cost baseline.

Some Important Terminology:

Project Budget

You must understand the relationship between contingency reserve, management reserve, and the project budget. These are crucial concepts; without these reserves, you cannot estimate the cost baseline and project budget.

Cost Estimate

The cost estimate is the cost of all work packages and is “rolled up” to the top level; this is the total cost of the project.

Some Important Terminology:

Cost Baseline

When you add the contingency reserve to the cost estimate, you get the cost baseline.

$$\text{Cost Baseline} = \text{Cost Estimate} + \text{Contingency Reserve}$$

Note that the project's performance will be measured against the cost baseline.

If you add the management reserve to the cost baseline, you will get the project budget.

$$\text{Project Budget} = \text{Cost Baseline} + \text{Management Reserve}$$

The Difference between Contingency Reserve and Management Reserve

The following are a few differences between contingency reserve and management reserve:

- ❑ The contingency reserve is used to manage identified risks, while management reserve is used to manage unidentified risks.
- ❑ The contingency reserve is an estimated figure. While Management reserve is a percentage of the cost or duration of the project.
- ❑ The project manager has authority over the contingency reserve. While For management reserve, they need management's permission.
- ❑ The contingency reserve is a part of the performance measurement baseline while management reserve is not.

Project Budget Template

Project Budget	Labor		Estimated time	Materials		Planned budget	Actual expenses	Difference between budget and expenses
	Worker	Hourly rate		No of units	Price / Unit			
Task 1	Lisa Johnson	\$45	30h	50	\$20	\$2,350.00	\$3,000.00	650
Task 2	John Bautista	\$45	70h	100	\$30	\$6,150.00	\$6,000.00	150
Task 3	Katie Stark	\$50	50h	20	\$40	\$3,300.00	\$3,500.00	200
Task 4	Austin White	\$40	120h	30	\$40	\$6,000.00	\$6,000.00	0
Task 5	Thomas Clark	\$50	30h	40	\$20	\$2,300.00	\$2,000.00	300